

Supplementary Material for “From Apparent Edge to No-Go: A Point-in-Time, Cost-Aware Audit of Heterogeneous XAUUSD Trading Strategies”

Supplementary Material^a

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ABSTRACT

This supplement reports the complete locked specification registry, detailed source-native gates, all primary specification outcomes, all familywise decisions, cross-feed results, timestamp-alignment sensitivity, deflated-Sharpe diagnostics, protocol deviations, audit scope, and evidence map. Machine-readable versions of all reported tables accompany the submission package.

S1. Registry construction and status labels

The registry was frozen before consolidated specification outcome computation. Every row identifies the source block, source candidate, economic family, analysis role, direction, horizon, reference interval, alignment and cross-feed requirements, source-code digest, configuration digest, and serialized signal contract. Exact file and registry digests are stored in the accompanying machine-readable manifest.

The 39 rows labeled PRIMARY_LOCKED_RERUN form the article’s primary universe. The 8 machine-learning rows are labeled HISTORICAL_FORENSIC_CASE_STUDY_ONLY; they were historically selected and had been exposed to later diagnostic information. They were intentionally not rerun and do not enter any primary result.

Table 1: Complete locked registry.

ID	Source block	Candidate	Economic family	Role
A1R-001	successor scan	trend continuation 12h fast	trend continuation	Primary
A1R-002	successor scan	trend continuation 48h medium	trend continuation	Primary
A1R-003	successor scan	trend continuation 72h slow	trend continuation	Primary
A1R-004	successor scan	trend pullback 24h	trend pullback	Primary
A1R-005	successor scan	trend pullback 48h	trend pullback	Primary
A1R-006	successor scan	range breakout 12h 24	range breakout	Primary
A1R-007	successor scan	range breakout 48h 72	range breakout	Primary
A1R-008	successor scan	volatility expansion 12h	volatility expansion	Primary
A1R-009	successor scan	volatility expansion 48h	volatility expansion	Primary
A1R-010	successor scan	low vol range reversion 12h	low vol range reversion	Primary
A1R-011	successor scan	low vol range reversion 24h	low vol range reversion	Primary
A1R-012	successor scan	zscore reversion 12h	zscore reversion	Primary
A1R-013	successor scan	zscore reversion 24h	zscore reversion	Primary

Supplementary Material: XAUUSD Strategy Audit

ID	Source block	Candidate	Economic family	Role
A1R-014	successor scan	failed breakout reversion 12h	failed breakout reversion	Primary
A1R-015	successor scan	post event momentum 12h	post event momentum	Primary
A1R-016	successor scan	post event momentum 24h	post event momentum	Primary
A1R-017	successor scan	post event breakout 24h	post event breakout	Primary
A1R-018	successor scan	london window breakout 12h	session breakout	Primary
A1R-019	successor scan	new york window breakout 12h	session breakout	Primary
A1R-020	cross asset fixed	usd silver consensus long 4h	USD PRECIOUS CONSENSUS	Primary
A1R-021	cross asset fixed	usd silver consensus short 4h	USD PRECIOUS CONSENSUS	Primary
A1R-022	cross asset fixed	silver lead catchup long 4h	PRECIOUS RELATIVE VALUE	Primary
A1R-023	cross asset fixed	silver lag breakdown short 4h	PRECIOUS RELATIVE VALUE	Primary
A1R-024	cross asset fixed	yield usd relief long 12h	RATES USD ALIGNMENT	Primary
A1R-025	cross asset fixed	yield usd pressure short 12h	RATES USD ALIGNMENT	Primary
A1R-026	cross asset fixed	riskoff rates relief long 12h	RISK RATES DIVERGENCE	Primary
A1R-027	cross asset fixed	riskon rates pressure short 12h	RISK RATES DIVERGENCE	Primary
A1R-028	cross asset fixed	energy inflation long 12h	ENERGY INFLATION IMPULSE	Primary
A1R-029	cross asset fixed	energy disinflation short 12h	ENERGY INFLATION IMPULSE	Primary
A1R-030	cross asset fixed	dxy silver shock long 4h	DXY PRECIOUS SHOCK	Primary
A1R-031	cross asset fixed	dxy silver shock short 4h	DXY PRECIOUS SHOCK	Primary
A1R-032	cross asset event response	event usd silver continuation short 4h	EVENT USD SILVER CONTINUATION	Primary
A1R-033	cross asset event response	event usd silver continuation short 12h	EVENT USD SILVER CONTINUATION	Primary
A1R-034	cross asset event response	event usd silver continuation long 4h	EVENT USD SILVER CONTINUATION	Primary
A1R-035	cross asset event response	event usd silver continuation long 12h	EVENT USD SILVER CONTINUATION	Primary
A1R-036	cross asset event response	event gold lag short 4h	EVENT GOLD LAG	Primary
A1R-037	cross asset event response	event gold lag long 4h	EVENT GOLD LAG	Primary
A1R-038	stage178 ml	logistic / direction 6h	SUPERVISED DIRECTIONAL META MODEL	Forensic only
A1R-039	stage178 ml	logistic / direction 12h	SUPERVISED DIRECTIONAL META MODEL	Forensic only
A1R-040	stage178 ml	logistic / direction 24h	SUPERVISED DIRECTIONAL META MODEL	Forensic only
A1R-041	stage178 ml	logistic / triple barrier 24h	SUPERVISED DIRECTIONAL META MODEL	Forensic only
A1R-042	stage178 ml	hist gb / direction 6h	SUPERVISED DIRECTIONAL META MODEL	Forensic only

ID	Source block	Candidate	Economic family	Role
A1R-043	stage178 ml	hist gb / direction 12h	SUPERVISED DIRECTIONAL META MODEL	Forensic only
A1R-044	stage178 ml	hist gb / direction 24h	SUPERVISED DIRECTIONAL META MODEL	Forensic only
A1R-045	stage178 ml	hist gb / triple barrier 24h	SUPERVISED DIRECTIONAL META MODEL	Forensic only
A1R-046	canonical tsmom	MOP2012 12M 1M VOL SCALED XAUUSD BROKER PROXY	TIME SERIES MOMENTUM	Primary
A1R-047	precious metals rv	GOLD SILVER 252D DYNAMIC LOG SPREAD MEAN REVERSION	GOLD SILVER DYNAMIC LOG SPREAD MEAN REVERSION	Primary

S2. Data contract and source bindings

The primary intraday contract is 2015-01-02 07:00 UTC through, but not including, 2025-01-01 00:00 UTC. Cross-asset fixed and event-response blocks begin on 2016-01-01. The relative-value rule begins on 2012-01-01 and requires a 252-session formation sample. The monthly trend rule begins after a 12-month return lookback and a minimum of 120 past daily returns.

AMarkets server timestamps are converted under `EU_DST_GMT_OFFSET_PAIR`: subtract 120 minutes during standard time and 180 minutes during European daylight-saving time. The data-contract preflight recorded zero duplicate timestamps and zero invalid OHLC rows. The AMarkets alignment database, Dukascopy extended history, timestamp contract, official-event timestamps, and all additional inputs are identified by file digest in the frozen main-run manifest. Raw provider data are not included in this journal package.

S3. Exact strategy-class contracts

S3.1. Successor price-only rules

Nineteen non-overlapping rules cover ten families: trend continuation, trend pullback, range breakout, volatility expansion, session breakout, post-event momentum, post-event breakout, failed-breakout reversion, low-volatility range reversion, and z -score reversion. The registry supplies each fixed direction, horizon, threshold, and signal contract. The reference interval is 2015-01-02 07:00 UTC to 2025-01-01 00:00 UTC exclusive.

S3.2. Cross-asset fixed rules

The 12 rules form six economic families with 4- or 12-hour horizons. Decisions occur during 06:00–19:00 UTC on weekdays and exclude official-event blackout periods. Standardized features use a 1,440-row rolling window and at least 480 past observations. Signals within a candidate are non-overlapping.

S3.3. Event-response rules

The six rules use 4- or 12-hour horizons and enter only when an official event occurred 15–120 minutes before the decision. The official-event file is hash-bound. Standardized 15-minute reactions in XAUUSD, XAGUSD, and the dollar composite determine the locked directional conditions.

S3.4. Time-series momentum

The sign of the prior 12 completed monthly returns determines a long or short position for one month. Volatility is estimated from past daily returns using a 60-day exponentially weighted center of mass and 261-day annualization. The annual volatility target is 20%, absolute position is capped at 2.0, and turnover is charged on the complete monthly position change. The comparison benchmark is passive volatility-scaled long gold under the same risk and cost model.

S3.5. Gold-silver relative value

The rule estimates $\log(XAU) = \alpha + \beta \log(XAG)$ over 252 completed daily closes. It enters at the next synchronized open after $|z| \geq 2$, exits at the next open following a mean crossing, adverse $|z| \geq 4$, or 60-day maximum holding, and uses gross-normalized fixed entry weights. The rule is not a parameter scan; structural-break and cointegration quantities are diagnostic only.

S4. Cost formulas and source-native gates

For each directional trade, gross basis-point return is $10,000s(P_{out}/P_{in} - 1)$. Source-specific net returns subtract the applicable round-trip or turnover cost. Severe costs are primary. Fixed stress is a secondary comparison: 10 bps for one-leg intraday rules and each unit of monthly turnover, and 20 bps for the two-leg rule.

Table 2: Source-native reference gates.

Block	Complete conjunction
Successor scan	At least 40 trades; severe mean > 0 ; profit factor ≥ 1.15 ; block-bootstrap 10th percentile > 0 ; positive mean after removing the largest winner; positive-fold share ≥ 0.60 ; locked-risk CAGR $\geq 2\%$.
Cross-asset fixed	At least 80 trades; severe mean > 0 ; profit factor ≥ 1.20 ; bootstrap 10th percentile > 0 ; positive after removing the largest winner; matched year/hour excess mean and bootstrap lower bound > 0 ; at least 2 of 3 positive folds; worst-fold mean ≥ -5 bps; maximum positive-year profit share $\leq 45\%$; locked-risk CAGR $\geq 2\%$.
Event response	At least 40 trades; severe mean > 0 ; profit factor ≥ 1.20 ; bootstrap 10th percentile > 0 ; positive after removing the largest winner; matched excess mean and lower bound > 0 ; at least 2 of 3 positive folds; worst-fold mean ≥ -5 bps; maximum positive-year profit share $\leq 50\%$; locked-risk CAGR $\geq 1\%$.
Canonical TSMOM	At least 120 months; severe annualized return $> 2\%$; Sharpe > 0.20 ; moving-block bootstrap 10th-percentile monthly mean > 0 ; annualized return after removing best month > 0 ; at least 2 of 3 positive folds; worst-fold annualized return $> -10\%$; positive-year share $\geq 55\%$; positive-year concentration $\leq 50\%$; maximum drawdown $\leq 50\%$; paired excess mean and bootstrap lower bound versus passive volatility-scaled gold > 0 .
Precious-metals RV	At least 25 trades; severe mean > 0 ; profit factor ≥ 1.20 ; bootstrap lower bound and remove-best-trade mean > 0 ; annualized return $\geq 2\%$; maximum drawdown $\leq 25\%$; at least 2 positive folds; worst-fold annualized return $\geq -5\%$; positive-year share $\geq 45\%$; year concentration $\leq 50\%$; trade concentration $\leq 25\%$; mean better than the inverted rule; random-side percentile ≥ 0.90 .

S5. Complete primary specification results

Table 3 contains all primary rows. The adjusted probability is the specification-level value obtained from the relevant within-family max- t distribution. The family decision uses the family maximum and is reported in Section S6.

Supplementary Material: XAUUSD Strategy Audit

Table 3: Complete primary specification results. Returns are mean basis points per native trade or rebalance.

ID	N	Gross	Normal	Severe	Fixed	Adj. <i>p</i>	Native pass
A1R-001	3476	-0.610	-3.610	-5.110	-10.610	1.000	No
A1R-002	865	2.995	-0.005	-1.505	-7.005	0.892	No
A1R-003	906	1.276	-1.724	-3.224	-8.724	0.960	No
A1R-004	1859	2.013	-0.987	-2.487	-7.987	0.948	No
A1R-005	908	0.805	-2.195	-3.695	-9.195	0.898	No
A1R-006	1702	-1.149	-4.149	-5.649	-11.149	1.000	No
A1R-007	455	7.146	4.146	2.646	-2.854	0.479	No
A1R-008	1932	-0.077	-3.077	-4.577	-10.077	1.000	No
A1R-009	649	5.813	2.813	1.313	-4.187	0.565	No
A1R-010	579	-1.900	-4.900	-6.400	-11.900	0.997	No
A1R-011	243	-2.679	-5.679	-7.179	-12.679	0.956	No
A1R-012	1706	-0.723	-3.723	-5.223	-10.723	1.000	No
A1R-013	717	-0.818	-3.818	-5.318	-10.818	0.989	No
A1R-014	1949	-2.995	-5.995	-7.495	-12.995	1.000	No
A1R-015	345	-2.633	-5.633	-7.133	-12.633	0.999	No
A1R-016	343	0.881	-2.119	-3.619	-9.119	0.898	No
A1R-017	164	-5.074	-8.074	-9.574	-15.074	0.930	No
A1R-018	1230	1.883	-1.117	-2.617	-8.117	0.989	No
A1R-019	1082	-0.067	-3.067	-4.567	-10.067	1.000	No
A1R-020	950	1.319	-1.681	-4.681	-8.681	1.000	No
A1R-021	1032	3.100	0.100	-2.900	-6.900	0.999	No
A1R-022	582	0.024	-2.976	-5.976	-9.976	1.000	No
A1R-023	532	3.851	0.851	-2.149	-6.149	0.979	No
A1R-024	525	2.335	-0.665	-3.665	-7.665	0.993	No
A1R-025	554	1.991	-1.009	-4.009	-8.009	0.992	No
A1R-026	330	1.721	-1.279	-4.279	-8.279	0.993	No
A1R-027	293	1.553	-1.447	-4.447	-8.447	0.991	No
A1R-028	308	3.959	0.959	-2.041	-6.041	0.914	No
A1R-029	318	0.159	-2.841	-5.841	-9.841	0.993	No
A1R-030	569	-1.058	-4.058	-7.058	-11.058	1.000	No
A1R-031	591	2.983	-0.017	-3.017	-7.017	0.992	No
A1R-032	226	-2.162	-5.162	-8.162	-12.162	1.000	No
A1R-033	154	0.520	-2.480	-5.480	-9.480	0.998	No
A1R-034	243	-1.795	-4.795	-7.795	-11.795	1.000	No
A1R-035	154	4.012	1.012	-1.988	-5.988	0.980	No
A1R-036	12	8.007	5.007	2.007	-1.993	0.652	No
A1R-037	21	7.542	4.542	1.542	-2.458	0.643	No
A1R-046	154	11.644	9.528	7.412	4.591	0.446	No
A1R-047	27	-2.403	-10.403	-18.403	-22.403	0.602	No

S6. Complete familywise decisions

All family tests use centered, synchronized, circular moving blocks of 20 calendar days, 5,000 replications, and seed 2026090201. Severe-cost returns are assigned to the UTC exit date and inactive dates receive zero. All 20 final decisions are NO_EDGE_UNDER_LOCKED_PROTOCOL.

Table 4: Complete familywise decisions.

Economic family	Specs.	Native passes	Best ID	Adj. <i>p</i>	Decision
DXY PRECIOUS SHOCK	2	0	A1R-031	0.992	No edge
ENERGY INFLATION	2	0	A1R-028	0.914	No edge
IMPULSE					
EVENT GOLD LAG	2	0	A1R-037	0.643	No edge
EVENT USD SILVER	4	0	A1R-035	0.980	No edge
CONTINUATION					
GOLD SILVER DYNAMIC	1	0	A1R-047	0.602	No edge
LOG SPREAD MEAN					
REVERSION					
PRECIOUS RELATIVE	2	0	A1R-023	0.979	No edge
VALUE					
RATES USD ALIGNMENT	2	0	A1R-025	0.992	No edge
RISK RATES DIVERGENCE	2	0	A1R-027	0.991	No edge
TIME SERIES MOMENTUM	1	0	A1R-046	0.446	No edge
USD PRECIOUS	2	0	A1R-021	0.999	No edge
CONSENSUS					
failed breakout reversion	1	0	A1R-014	1.000	No edge

Economic family	Specs.	Native passes	Best ID	Adj. p	Decision
low vol range reversion	2	0	A1R-011	0.956	No edge
post event breakout	1	0	A1R-017	0.930	No edge
post event momentum	2	0	A1R-016	0.898	No edge
range breakout	2	0	A1R-007	0.479	No edge
session breakout	2	0	A1R-018	0.989	No edge
trend continuation	3	0	A1R-002	0.892	No edge
trend pullback	2	0	A1R-005	0.898	No edge
volatility expansion	2	0	A1R-009	0.565	No edge
zscore reversion	2	0	A1R-013	0.989	No edge

S7. Cross-feed replication

Replication is applicable to 19 successor rules and the canonical trend rule. Passing requires at least 30 matched outcomes, signal Jaccard similarity ≥ 0.60 , matched gross-return correlation ≥ 0.90 , and consistent severe-cost effect direction. Five failures are Jaccard failures and two are sign reversals.

Table 5: Complete cross-feed results for eligible specifications.

ID	Primary N	Secondary N	Jaccard	Return corr.	Secondary severe	Pass
A1R-001	3476	3546	0.633	0.999	-5.186	Yes
A1R-002	865	880	0.493	1.000	-1.927	No
A1R-003	906	917	0.403	1.000	-2.881	No
A1R-004	1859	1897	0.516	1.000	-2.127	No
A1R-005	908	915	0.393	0.999	-4.121	No
A1R-006	1702	1768	0.825	1.000	-5.008	Yes
A1R-007	455	462	0.743	1.000	-0.503	No
A1R-008	1932	2023	0.825	1.000	-4.519	Yes
A1R-009	649	680	0.765	1.000	-0.406	No
A1R-010	579	614	0.601	1.000	-5.495	Yes
A1R-011	243	258	0.561	1.000	-5.832	No
A1R-012	1706	1756	0.731	1.000	-4.823	Yes
A1R-013	717	734	0.738	1.000	-5.488	Yes
A1R-014	1949	2027	0.709	1.000	-7.103	Yes
A1R-015	345	352	0.910	1.000	-7.735	Yes
A1R-016	343	355	0.902	1.000	-3.849	Yes
A1R-017	164	174	0.888	1.000	-7.962	Yes
A1R-018	1230	1304	0.850	1.000	-1.926	Yes
A1R-019	1082	1105	0.941	1.000	-4.255	Yes
A1R-046	106	106	1.000	0.998	-5.005	Yes

S8. Timestamp-alignment warning sensitivity

Trades overlapping 2019-10-07 through 2019-10-14 are excluded. No applicable specification changes severe-cost effect sign or source-native pass status, and no family changes multiplicity rejection or support label.

Table 6: Complete alignment-warning sensitivity results for applicable specifications.

ID	Primary N	Sensitivity N	Primary severe	Sensitivity severe
A1R-001	3476	3470	-5.110	-5.081
A1R-002	865	864	-1.505	-1.540
A1R-003	906	903	-3.224	-2.968
A1R-004	1859	1856	-2.487	-2.413
A1R-005	908	906	-3.695	-3.583
A1R-006	1702	1701	-5.649	-5.653
A1R-007	455	455	2.646	2.646
A1R-008	1932	1930	-4.577	-4.632
A1R-009	649	648	1.313	1.481
A1R-010	579	579	-6.400	-6.400
A1R-011	243	243	-7.179	-7.179
A1R-012	1706	1706	-5.223	-5.223
A1R-013	717	717	-5.318	-5.318
A1R-014	1949	1945	-7.495	-7.540
A1R-015	345	343	-7.133	-7.021
A1R-016	343	341	-3.619	-3.689
A1R-017	164	163	-9.574	-10.231

Supplementary Material: XAUUSD Strategy Audit

ID	Primary N	Sensitivity N	Primary severe	Sensitivity severe
A1R-018	1230	1228	-2.617	-2.559
A1R-019	1082	1081	-4.567	-4.548
A1R-020	950	950	-4.681	-4.681
A1R-021	1032	1031	-2.900	-2.880
A1R-022	582	580	-5.976	-6.006
A1R-023	532	530	-2.149	-2.140
A1R-024	525	524	-3.665	-3.778
A1R-025	554	554	-4.009	-4.009
A1R-026	330	329	-4.279	-4.407
A1R-027	293	293	-4.447	-4.447
A1R-028	308	308	-2.041	-2.041
A1R-029	318	318	-5.841	-5.841
A1R-030	569	569	-7.058	-7.058
A1R-031	591	591	-3.017	-3.017
A1R-032	226	224	-8.162	-8.014
A1R-033	154	152	-5.480	-5.301
A1R-034	243	243	-7.795	-7.795
A1R-035	154	154	-1.988	-1.988
A1R-036	12	12	2.007	2.007
A1R-037	21	21	1.542	1.542

S9. Deflated Sharpe ratios and PBO status

The DSR uses a common daily calendar from 2016-01-01 through 2024-12-31, 365-day annualization, 39 eligible trials, and a 0.95 reporting threshold. The expected maximum daily Sharpe is 0.0477359441. No specification passes.

Table 7: Complete deflated-Sharpe diagnostic.

ID	Daily Sharpe	Annualized	Skewness	Kurtosis	DSR probability	Pass
A1R-001	-0.0782	-1.495	0.011	14.682	0.00000	No
A1R-002	-0.0029	-0.055	-0.487	31.929	0.00185	No
A1R-003	-0.0145	-0.277	-0.686	24.695	0.00017	No
A1R-004	-0.0261	-0.498	-0.784	17.008	0.00001	No
A1R-005	-0.0174	-0.332	-1.434	33.081	0.00008	No
A1R-006	-0.0511	-0.976	-1.192	42.525	0.00000	No
A1R-007	0.0081	0.155	1.137	57.077	0.01128	No
A1R-008	-0.0452	-0.864	0.538	19.590	0.00000	No
A1R-009	-0.0054	-0.104	-0.125	23.001	0.00115	No
A1R-010	-0.0314	-0.601	0.035	54.192	0.00000	No
A1R-011	-0.0207	-0.396	-0.852	140.374	0.00004	No
A1R-012	-0.0574	-1.097	0.049	44.963	0.00000	No
A1R-013	-0.0323	-0.616	-0.547	30.207	0.00000	No
A1R-014	-0.0817	-1.561	-0.098	17.677	0.00000	No
A1R-015	-0.0264	-0.504	1.610	79.141	0.00002	No
A1R-016	-0.0073	-0.140	2.250	61.477	0.00088	No
A1R-017	-0.0258	-0.493	-0.916	99.793	0.00001	No
A1R-018	-0.0123	-0.235	0.400	15.993	0.00030	No
A1R-019	-0.0404	-0.772	-0.097	28.749	0.00000	No
A1R-020	-0.0608	-1.161	0.296	24.785	0.00000	No
A1R-021	-0.0348	-0.664	3.243	50.181	0.00000	No
A1R-022	-0.0566	-1.081	-0.562	51.631	0.00000	No
A1R-023	-0.0192	-0.368	5.580	131.678	0.00014	No
A1R-024	-0.0230	-0.440	-0.282	35.432	0.00002	No
A1R-025	-0.0223	-0.427	-0.980	62.268	0.00003	No
A1R-026	-0.0227	-0.434	-0.037	50.483	0.00003	No
A1R-027	-0.0216	-0.412	0.923	79.536	0.00004	No
A1R-028	-0.0087	-0.166	0.765	88.647	0.00064	No
A1R-029	-0.0230	-0.439	-1.797	120.800	0.00002	No
A1R-030	-0.0619	-1.183	-1.462	40.979	0.00000	No
A1R-031	-0.0235	-0.449	1.196	70.973	0.00003	No
A1R-032	-0.0380	-0.726	-1.058	77.592	0.00000	No
A1R-033	-0.0158	-0.302	-5.519	222.002	0.00008	No
A1R-034	-0.0415	-0.793	-5.036	100.119	0.00000	No
A1R-035	-0.0075	-0.144	-1.046	107.494	0.00074	No
A1R-036	0.0024	0.047	17.201	790.882	0.00401	No
A1R-037	0.0028	0.054	13.689	800.348	0.00435	No
A1R-046	-0.0046	-0.088	0.174	98.651	0.00136	No
A1R-047	-0.0053	-0.102	3.422	318.589	0.00130	No

PBO is not estimable because the universe does not form a valid combinatorially symmetric cross-validation matrix: event-only, intraday, monthly, and two-leg strategies have heterogeneous calendars, holding horizons, and objectives. No substitute scalar was computed after the outcome.

S10. Protocol deviations

A1-DEV-001. The redacted evidence copy of the canonical trend script had a different digest. Independent package evidence established that three malformed redaction placeholders in authorization and decision-reporting statements had been restored. No signal-generation or performance-computation statement changed. The registry was reissued as version 1.1 before the consolidated rerun.

A1-DEV-002. The first unified run requested the Dukascopy H1 table from the AMarkets SQLite file and aborted before entering any specification-outcome loop. Version 1.1 routed the AMarkets and Dukascopy tables to their already hash-bound databases and added schema gates. No strategy, cost, statistic, or seed changed. The aborted directory was retained.

S11. Independent audit boundary

The independent audit verified safe archive paths, no duplicate members, CRC integrity, manifest completeness, file byte sizes and hashes, lock hashes, registry identities, outcome-table uniqueness, reference boundaries, finite returns, and the 28,685-row count. It independently recomputed each specification's count, four mean-return layers, severe-cost profit factor, all family max- t results, and all DSR probabilities.

The audit did not regenerate every signal and trade from raw broker/vendor databases. It is therefore an output-level statistical reproduction with strong provenance checks, not a complete independent raw-data replication.

S12. Evidence ledger

Table 8: Central claim-to-artifact map.

Claim	Authoritative file	Field or check
47 registered; 39 primary; 8 forensic-only	article1_reference_rerun_summary.json	Registry counts
20 families; none supported	family_reference_decisions.csv	decision
28,685 normalized rows	all_primary_reference_trades.csv.gz	Independent row count
Positive counts 24/9/5/1	specification_reference_metrics.csv	Four mean-return columns
No native pass	specification_reference_metrics.csv	source_native_pass
Minimum family $p = 0.446111$	family_reference_decisions.csv	family_adjusted_p
Cross-feed: 13 pass, 7 fail	cross_feed_replication.csv	Applicability and checks
No alignment fragility	alignment_warning_sensitivity.csv	Sign/pass changes
Maximum DSR probability 0.011284	deflated_sharpe_ratio.csv	DSR probability
PBO not estimable	evaluation_protocol.json	Status and reason

Claim	Authoritative file	Field or check
Audit passed with limitations	<code>independent_audit_report.json</code>	Integrity and recomputation

S13. Reproduction instructions

Run `python3 build_figures.py` in a directory containing the supplied CSV files to reproduce all figures and generated LaTeX tables. Compile each TeX entry point with `latexmk -pdf`. The article-level package contains the CAS single-column class and bibliography style at the same directory level because Elsevier's Editorial Manager does not process nested LaTeX source folders.